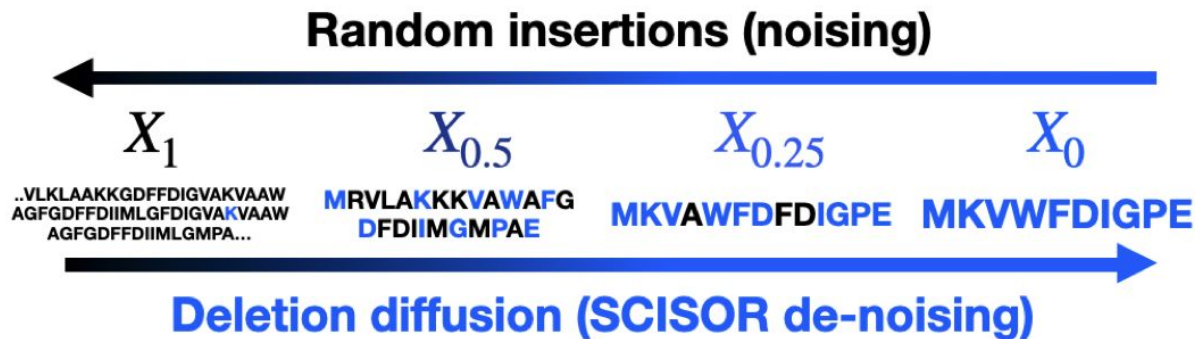


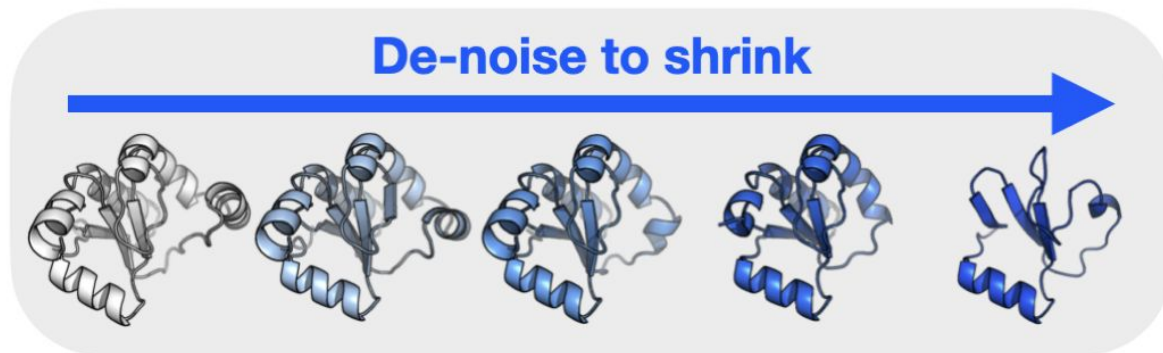
A Diffusion Model to Shrink Proteins While Maintaining their Function

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SCISOR



(a)



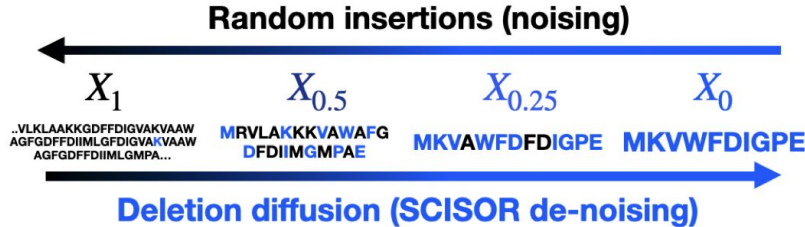
Motivation

- Lately, many more proteins are designed and synthesized in the lab.
- Longer protein -> Higher cost as well as complications
- For shortening a protein sequence -> look for shorter homologues, or do expensive simulations.
- Instead -> We can **learn** to shrink from millions of sequences

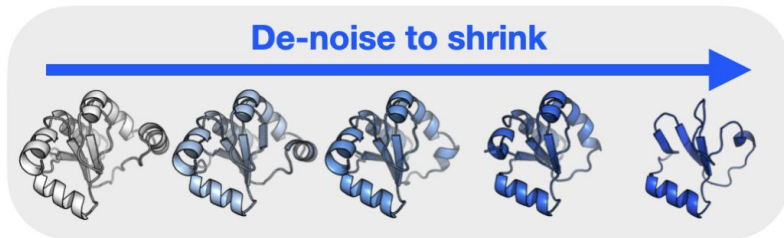
Proposed Model: SCISOR

Sequence Contraction with InSersion-Only noising pRocess

- Generate (noisy) longer sequences via random insertions
- Train diffusion model to “denoise” by shrinking to its natural state.



(a)



Forward Process: Noising

Based on “pure birth” process (continuous-time markov chain). Assume a “insertion rate” $B(t)$, where insertion letters pulled from distribution $\text{Cat}(\pi)$.

Algorithm 1 Sample X_t

Require: Initial sequence $X = X^{(1)} \dots X^{(L)}$, time t

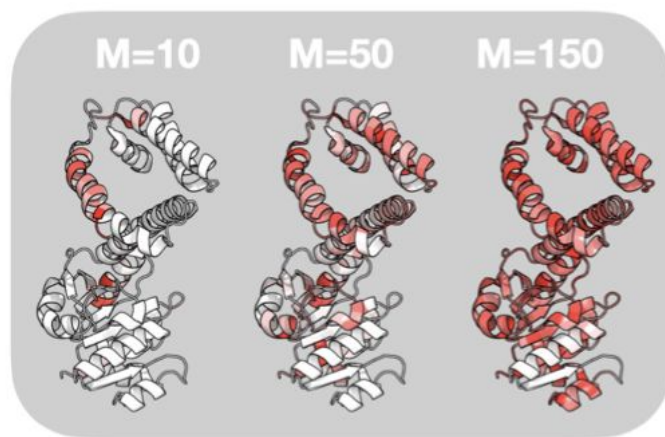
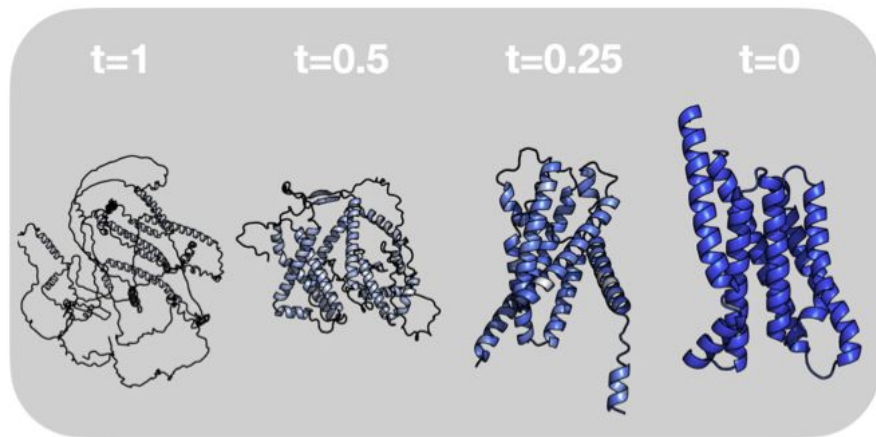
- 1: Compute the probability of no insertions at a site $\alpha(t) \leftarrow \exp \left(- \int_0^t \beta(s) ds \right)$
 - 2: Sample total number of insertions up to time t , $M_t \sim \text{NegativeBinomial}(L + 1, \alpha(t))$
 - 3: Sample the number of insertions in each position by uniformly distributing M_t into $L + 1$ bins:
 $(\ell_0, \dots, \ell_L) \sim \text{UniformMultinomial}(M_t)$
 - 4: **for** $j = 0$ to L **do**
 - 5: Sample insertion Y_j of length ℓ_j , with each character independently from $\text{Cat}(\pi)$
 - 6: **end for**
 - 7: Add insertions into X to construct $X_t \leftarrow Y_0 X^{(0)} Y_1 X^{(1)} \dots X^{(L)} Y_L$
 - 8: **return** X_t
-

Reverse Process: De-Noising

Given a noisy sequence X and total number of insertions M , learn

$$q_{\theta}(\text{prev}(X) \mid X, M)$$

Here $\text{prev}(X)$ is the last letter that was inserted to X . Thus, we learn to predict latest “noise letter”.



Loss (for the interested)

Proposition 3.2. (Proof in App. [G](#)) Define M_t as the number of mutations up to time t , and $\text{prev}(X_t)$ is the last sequence that gained an insertion to become X_t . Then the negative log likelihood of a sequence of length L , $-\log q_\theta(X_0|L)$, is smaller than

$$\begin{aligned} & \mathbb{E}_{M_1} \text{KL}(p(X_1 | X_0, M_1) || q(X_1|L + M_1)) \\ & + \mathbb{E}_{t, X_t, M_t} \frac{M_t \beta(t)}{1 - \alpha(t)} \text{KL}(p(\text{prev}(X_t) | X_0, X_t, M_t) || q_\theta(\text{prev}(X_t) | X_t, M_t)) \end{aligned} \quad (3)$$

The first term is the quantity in Eqn. [2](#) – how well we can approximate $p(X_1)$; it is small as long as M_1 is typically large, i.e. $\alpha(1)$ is small, and can be calculated as in App. [C](#). The second term is the quantity we use to train the de-noiser. q_θ takes in X_t and the number of insertions in X_t and must predict which letter of X_t was last inserted – $\text{prev}(X_t)$. To train the model, we must calculate $p(\text{prev}(X_t)|X_0, X_t, M_t)$.

One Problem for Training -> How to decide last mutation

We know X_0 and X_t but not $\text{prev}(X_t)$.

However, we'd like to learn $p(\text{prev}(X_t) | X_0, X_t, M_t)$.

Idea: Letters that align with X_0 less often, are more likely to have been $\text{prev}(X_t)$

Proposition 3.3. (Proof in App. G) Call $\text{ali}(X, Y)$ the number of ways to align a sequence X to a sequence Y . Call b the letter that was deleted from X_t to $\text{prev}(X_t)$.

$$p(\text{prev}(X_t) | X_0, X_t, M_t) = \frac{\text{ali}(X_0, \text{prev}(X_t))}{M_t \cdot \text{ali}(X_0, X_t)}.$$

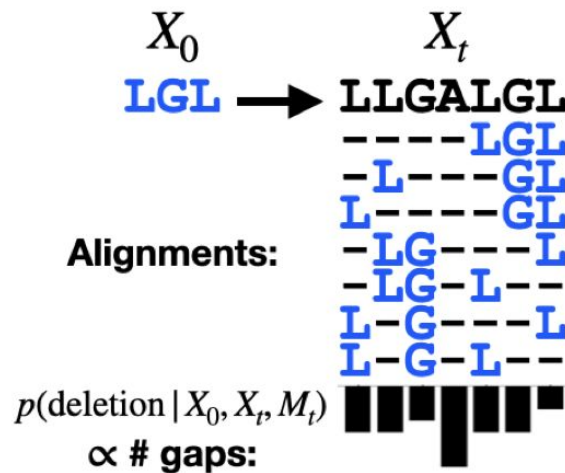


Figure 2: To calculate our target distribution of what letter to delete, $p(\text{prev}(X_t) | X_0, X_t, M_t)$, we align our starting sequence X_0 to our “noised” sequence X_t . The reverse process assigns a higher probability to deleting letters that are gaps in more of the alignments.

How to Use In Practice: Shrinking

Algorithm 2 Conditional shrinking of a sequence

Require: Number of deletions M , initial sequence X , temperature T .

- 1: Initialize $\tilde{X} \leftarrow X$
 - 2: **while** $|\tilde{X}| > L - M$ **do**
 - 3: Remove one letter from $\tilde{X}$ according to $q_{\theta}(\text{prev}(\tilde{X}) \mid \tilde{X}, M)^{1/T}$
 - 4: $M \leftarrow M - 1$
 - 5: **end while**
 - 6: **return** $\tilde{X}$
-

How to Use In Practice: Generation

Algorithm 3 Unconditional sequence generation with corrector steps

Require: Desired sequence length L , corrector steps K .

- 1: Sample $M \sim \text{NegativeBinomial}(L + 1, \alpha(1))$
 - 2: Sample X of length $L + M$ where each $X^{(j)} \sim \text{Cat}(\pi)$ independently
 - 3: **while** $|X| > L$ **do**
 - 4: **for** $k = 1, \dots, K$ **do** ▷ Corrector steps
 - 5: Remove one letter from X according to $q_\theta(\text{prev}(X) \mid X, M)$
 - 6: Insert a random letter from the distribution π into a random position in X
 - 7: **end for**
 - 8: Remove one letter from X according to $q_\theta(\text{prev}(X) \mid X, M)$
 - 9: $M \leftarrow M - 1$
 - 10: **end while**
 - 11: **return** X
-

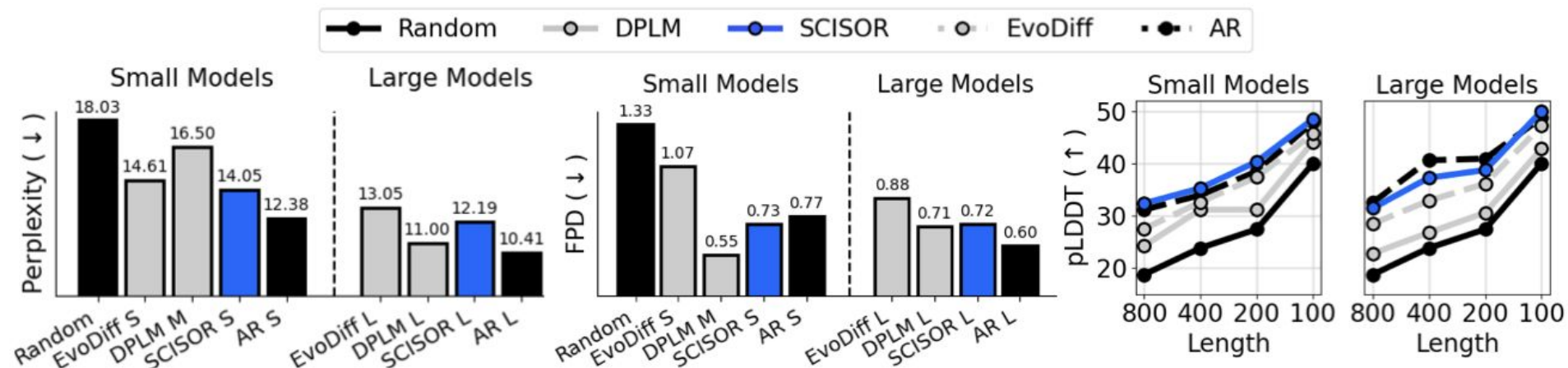
Results: Natural Sequence Generation

Same training-test split as EvoDiff (March 2020 release of Uniref50).

EvoDiff, DPLM are diffusion models (S M L refer to model size)

AR is autoregressive model.

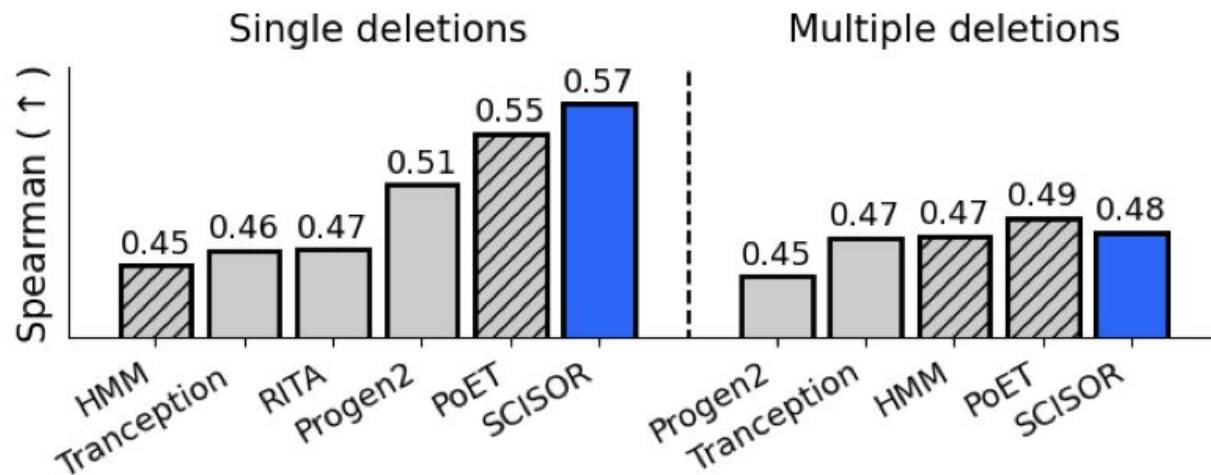
Perplexity, Fréchet protein distance(FPD), pLDDT



Results: Deletion Effect Prediction

ProteinGYM dataset. Spearman correlation.

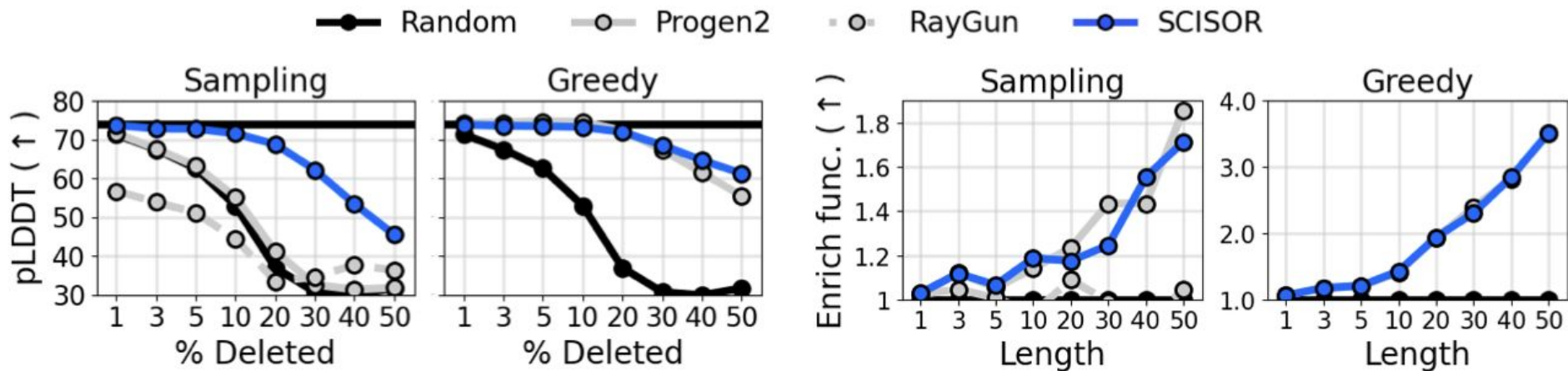
Shaded bars are models that use MSA information.



Results: Shrinking

First two plots: Foldability after shrinking.

Last two plots: Relative enrichment of non-interrupted active/binding sites.



Conclusion

- A diffusion model that is capable of naturally shrinking proteins as well as generation of new proteins.
- Results imply success to some extent.

Limitations

- Insertion process is not natural. Shrinking probabilities are based on it.
- Model supports only deletions, no substitutions or insertions is incorporated into the process.
- Shrinking process is guided towards only “natural proteins” without any other conditioning. This aspect may be further explored.

Supplementary Results

Table 1: ProteinGym results on single and multiple deletions.

Model	MSA	Single Deletions	Multiple Deletions
Progen2 S		0.457	0.445
Progen2 M		0.513	0.385
Progen2 Base		0.497	0.408
Progen2 L		0.491	0.375
Progen2 XL		0.393	0.392
RITA S		0.409	0.274
RITA M		0.448	0.318
RITA L		0.465	0.323
RITA XL		0.440	0.161
Tranception S		0.439	0.475
Tranception M		0.464	0.424
Tranception L		0.445	0.426
HMM	Yes	0.453	0.474
PoET (200M)	Yes	<u>0.551</u>	0.488
SCISOR S		0.332	0.241
SCISOR M		0.505	<u>0.478</u>
SCISOR L		0.573	<u>0.458</u>

Supplementary Results

Table 2: ProteinGym results on single deletions stratified by the measured function of each assay.

Model	MSA	Activity	Expression	Organismal Fitness	Stability
Progen2 S		0.566	0.294	0.499	0.470
Progen2 M		0.574	0.404	0.558	0.514
Progen2 Base		0.592	0.380	0.496	0.520
Progen2 L		0.550	0.344	0.560	0.508
Progen2 XL		0.418	0.298	0.333	0.521
RITA S		0.507	0.320	0.452	0.356
RITA M		0.514	0.345	0.500	0.432
RITA L		0.530	0.437	0.420	0.474
RITA XL		0.532	0.385	0.360	0.481
Tranception S		0.542	0.351	0.532	0.331
Tranception M		0.594	0.340	0.526	0.395
Tranception L		0.533	0.336	0.445	0.466
HMM	Yes	0.496	0.321	0.501	0.493
PoET (200M)	Yes	0.664	0.424	0.566	0.551
SCISOR S		0.376	0.289	0.198	0.465
SCISOR M		0.514	0.362	<u>0.576</u>	<u>0.571</u>
SCISOR L		<u>0.604</u>	<u>0.415</u>	0.668	0.606

Supplementary Results

Table 3: ProteinGym results on single deletions stratified by the MSA depth of proteins in each assay.

Model	MSA	Low	Medium	High
Progen2 S		0.558	0.429	0.497
Progen2 M		0.415	0.483	0.544
Progen2 Base		0.438	0.460	0.568
Progen2 L		0.513	0.473	0.532
Progen2 XL		0.216	0.499	0.530
RITA S		0.300	0.293	0.424
RITA M		0.278	0.376	0.492
RITA L		0.444	0.434	0.504
RITA XL		0.139	0.462	0.508
Tranception S		0.467	0.316	0.360
Tranception M		0.297	0.358	0.447
Tranception L		0.519	0.391	0.518
HMM	Yes	<u>0.624</u>	0.506	0.471
PoET (200M)	Yes	0.595	<u>0.553</u>	0.548
SCISOR S		0.385	0.381	0.509
SCISOR M		0.641	0.547	<u>0.575</u>
SCISOR L		0.621	0.628	0.584

Supplementary Results

Table 4: ProteinGym results on single deletions stratified by the taxa of the protein in each assay.

Model	MSA	Human	Eukaryote	Prokaryote	Virus
Progen2 S		0.506	0.467	0.361	0.567
Progen2 M		0.536	0.539	0.432	0.510
Progen2 Base		0.568	0.541	0.396	0.471
Progen2 L		0.536	0.513	0.442	0.492
Progen2 XL		0.511	0.537	0.420	0.573
RITA S		0.398	0.353	0.272	0.448
RITA M		0.496	0.417	0.310	0.504
RITA L		0.522	0.473	0.327	0.568
RITA XL		0.510	0.466	0.386	0.555
Tranception S		0.332	0.363	0.297	0.449
Tranception M		0.443	0.406	0.295	0.468
Tranception L		0.511	0.462	0.348	0.513
HMM	Yes	<u>0.585</u>	0.392	0.437	0.547
PoET (200M)	Yes	0.554	0.522	0.523	0.721
SCISOR S		0.486	0.418	0.340	0.652
SCISOR M		0.571	<u>0.539</u>	<u>0.525</u>	<u>0.735</u>
SCISOR L		0.590	0.568	0.621	0.767

Thanks For Listening

